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Cite this: DOI: 10.1039/c8qo01202d

Efficient synthesis of tetrasubstituted 2,3-allenoates and preliminary studies on bioactivities†

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Received 6th November 2018,
Accepted 10th December 2018

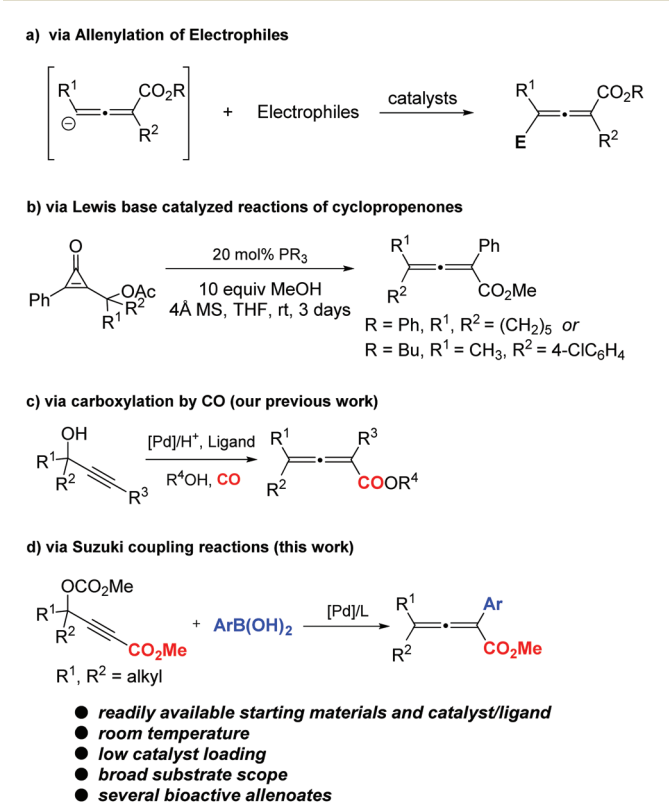
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A general and highly efficient straightforward protocol for the preparation of tetrasubstituted 2,3-allenoates through a palladium-catalyzed coupling reaction of 3-alkoxycarbonyl propargylic carbonates and boronic acids with commercially available tri(*o*-tolyl)phosphine as a ligand has been developed. This mild reaction exhibited a wide substrate scope and functional group compatibility, providing a highly efficient strategy to access valuable tetrasubstituted 2,3-allenoate products. Several tetrasubstituted 2,3-allenoates obtained in this study exhibited potent anti-cancer and anti-diabetic activities.

Allenes are of great importance not only because of their broad existence in natural products and biologically active compounds as well as advanced materials but also due to their versatility in organic synthesis as useful building blocks and even ligands/catalysts.[1–5] 2,3-Allenoates represent a particularly useful subset of allenes with demonstrated reactivities, including Morita–Baylis–Hillman reactions,[6] nucleophilic addition reactions,[7] electrophilic addition reactions,[8] rearrangements,[9] and cyclization reactions.[10] In the past few decades, substantial efforts have been devoted to the synthesis of 2,3-allenoates.[11] Wittig[11a,b] or Horner–Wadsworth–Emmons reaction,[11c] isomerization of alkynoates,[11d–f] β-hydride elimination of (*E*)-enol triflates,[11g,h] cross-coupling of α-diazoesters with alkynes,[11i,j] and caboxylation[11k–r] are the common approaches. However, limitations still remain for the synthesis of tetrasubstituted 2,3-allenoates: trapping of allenyl anions with electrophiles (Scheme 1a) developed by Hammond, Maruoka, and Feng *et al.*,[12a–d] the reaction of silyl alkynyl ketene acetals with electrophiles reported by List *et al.*,[12e] and phosphine-catalyzed ring opening reactions of cyclopropenonyl acetates observed by Shi *et al.* (Scheme 1b) may afford tetrasubstituted 2,3-allenoates.[12f] Recently, our laboratory reported the synthesis of tetrasubstituted 2,3-allenoates from readily available tertiary 2-alkynols (Scheme 1c).[11p]

During the past few decades, palladium-catalyzed coupling of propargylic derivatives with organometallic reagents has been used for the synthesis of allenes.[13] We envisioned to develop an efficient CO-free synthesis of tetrasubstituted 2,3-allenoates from readily available 3-alkoxycarbonyl-substituted propargylic carbonates and commercially available boronic acids *via* Suzuki coupling (Scheme 1d).



Scheme 1 Different protocols for the synthesis of tetrasubstituted 2,3-allenoates.

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†Electronic supplementary information (ESI) available. See DOI: 10.1039/c8qo01202d

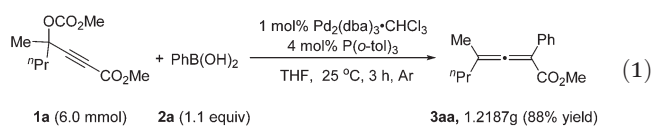
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In our initial studies, 3-methoxycarbonyl-substituted propargylic carbonate **1a** and phenylboronic acid **2a** were used as model substrates. When a mixture of **1a** and **2a** was treated with the semi-labile bidentate *o*-(diphenylphosphino)benzaldehyde (**L1**),[13w] only 2% yield of the desired 2,3-allenoate **3aa** was observed. Zheda-Phos **L2**[14] did not give any allenoate product. DPEPhos and *rac*-BINAP also failed in promoting the reaction efficiently. We were delighted to find that when the simplest phosphine ligand PPh_3 was employed in the reaction system, the desired product could be obtained smoothly with 53% yield. Interestingly, the commercially available ligand P(*o*-tol)$_3$[13x] was more efficient, affording allenoate **3aa** with 78% yield (Table 1).

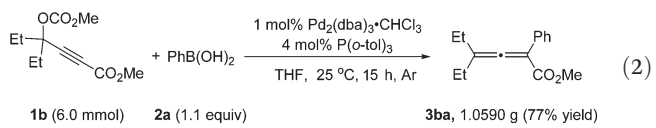
Other parameters were then investigated (Table 2). [Pd(allyl)Cl]$_2$ was much less efficient (entry 1). Although $Pd_2(dba)_3$, $Pd(OAc)_2$, and $Pd(PPh_3)_4$ could give the desired allenoate **3aa**, $Pd_2(dba)_3{\cdot}CHCl_3$ was still the most efficient (entries 2–5). Subsequently, different solvents were also surveyed. MeCN failed to provide better results, toluene afforded 87% yield, and the yield increased to 92% when THF was used (entries 6–8). We were pleased to find that the catalyst loading could be reduced to 1 mol%, and water was not necessary for this transformation even at 25 °C (entries 9 and 10) as compared to what was reported in our previous study.[13x]

With the established reaction conditions in hand, we next examined the reaction scope. As shown in Table 3, a wide range of 3-methoxycarbonyl-2-alkynylic carbonates **1** and boronic acids **2** may form the desired tetrasubstituted 2,3-allenoates **3** with excellent efficiency and regioselectivity (no alkyne products were observed in all cases). A variety of functional groups, such as phenol (**3ae**), nitro (**3af**), ester (**3ag**), aldehyde (**3ah**), and halides (**3ai** and **3cm**) survived under the standard conditions. The aryl boronic acids with the methoxy group at different positions all reacted smoothly to give the desired products **3bb–3bd**, although the reaction was slower when the methoxy group was at the *ortho*-position, probably due to steric hindrance (**3bd**). 2-Naphthyl or 3-thienyl boronic acid could also produce 2,3-allenoates **3ak** and **3al** with over 80% yields. It is worth mentioning that the bulky substituent in propargylic carbonate would not affect the efficiency (**3cm**). Synthetically useful functional groups, such as acetal (**3ea**) and the C=C bond (**3fa**) could be installed in the propargylic carbonates.

This protocol can be conducted well on a gram-scale: reaction of **1a** and **1b** afforded **3aa** and **3ba** in 88% and 77% yield, respectively (eqn (1) and (2)).



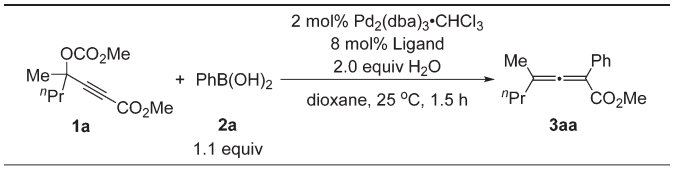
1a (6.0 mmol) **2a** (1.1 equiv) **3aa**, 1.2187g (88% yield) (1)



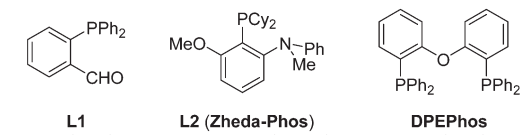
1b (6.0 mmol) **2a** (1.1 equiv) **3ba**, 1.0590 g (77% yield) (2)

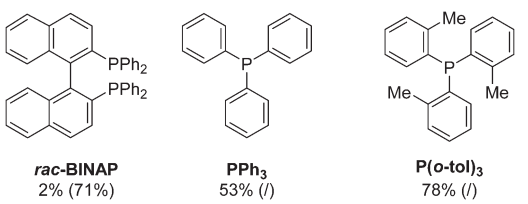
As useful intermediates in organic chemistry, the tetrasubstituted 2,3-allenoates can be converted to other versatile chemicals. As shown in Scheme 2, in the presence of $CuBr_2$, 4-bromo-2(5*H*)-furanone (**4**) could be obtained with 76% yield through bromolactonisation.[15] Treatment with Et_2Zn afforded

Table 1 Screening of ligands[a]



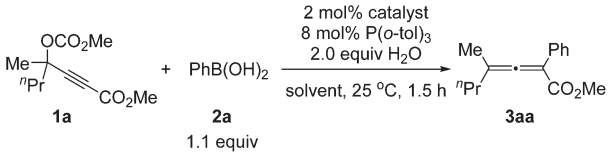
1a **2a** (1.1 equiv) **3aa**





[a] The reaction was conducted with 0.25 mmol of **1a**, 1.1 equiv. of **2a**, 2 mol% of $Pd_2(dba)_3{\cdot}CHCl_3$, 8 mol% of ligand, and 2.0 equiv. of H_2O in 0.5 mL of distilled dioxane at 25 °C for 1.5 h under a N_2 atmosphere. The yield and recovery were determined by ^{1}H NMR analysis, and the recovery of **1a** is shown in the parentheses.

Table 2 Optimization of the reaction conditions[a]



1a **2a** **3aa**

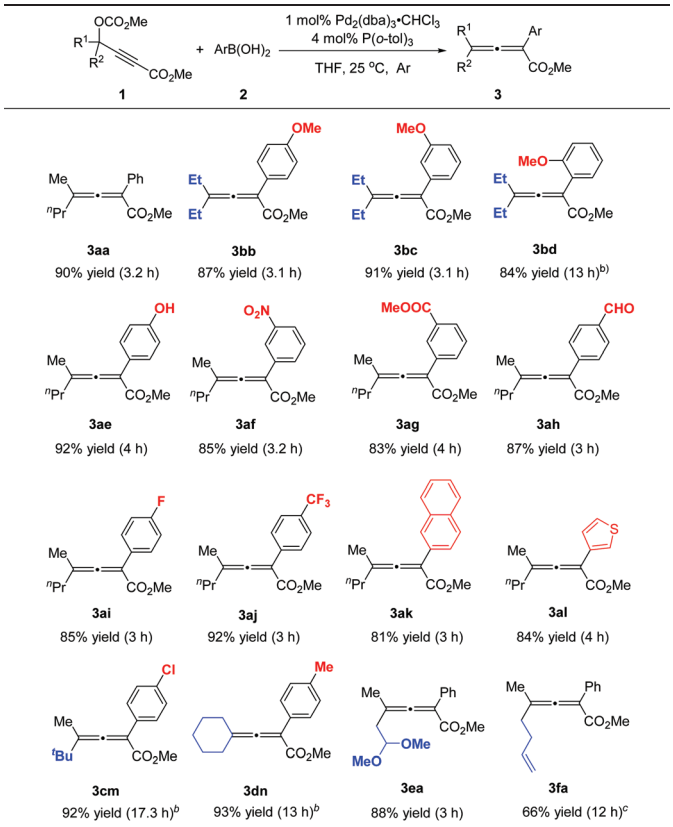
Entry	Catalyst	Solvent	Yield of **3aa** (%)	Recovery of **1a** (%)
1	$[Pd(allyl)Cl]_2$	Dioxane	7	90
2	$Pd_2(dba)_3$	Dioxane	63	—
3[b]	$Pd(OAc)_2$	Dioxane	69	—
4[c]	$Pd(PPh_3)_4$	Dioxane	72	—
5	$Pd_2(dba)_3{\cdot}CHCl_3$	Dioxane	78	—
6	$Pd_2(dba)_3{\cdot}CHCl_3$	MeCN	14	88
7	$Pd_2(dba)_3{\cdot}CHCl_3$	Toluene	87	—
8	$Pd_2(dba)_3{\cdot}CHCl_3$	THF	92	—
9[d]	$Pd_2(dba)_3{\cdot}CHCl_3$	THF	91	—
10[d,e]	**$Pd_2(dba)_3{\cdot}CHCl_3$**	**THF**	**92**	**—**

[a] The reaction was conducted with 0.25 mmol of **1a**, 1.1 equiv. of **2a**, 2 mol% of catalyst, 8 mol% of P(*o*-tol)$_3$, and 2.0 equiv. of H_2O in 0.5 mL of solvent at 25 °C for 1.5 h under a N_2 atmosphere. The yield and recovery were determined by ^{1}H NMR analysis. [b] 4 mol% of $Pd(OAc)_2$ was used. [c] 4 mol% of $Pd(PPh_3)_4$ was used. [d] The reaction was conducted with 1 mol% of $Pd_2(dba)_3{\cdot}CHCl_3$ and 4 mol% of P(*o*-tol)$_3$ for 3 h. [e] Without H_2O.

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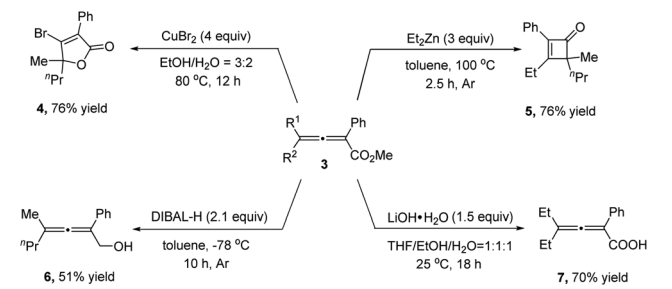
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Table 3 Substrate scope[a]



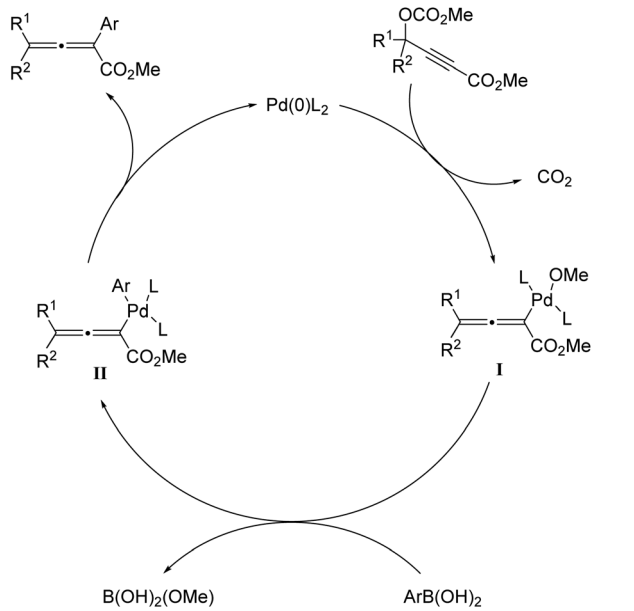
[a] The reaction was conducted with 1.0 mmol of **1**, 1.1 mmol of **2**, 1 mol% of $Pd_2(dba)_3{\cdot}CHCl_3$, and 4 mol% of P(*o*-tol)$_3$ in 2.0 mL of distilled THF at 25 °C under an Ar atmosphere. [b] 2 mol% of $Pd_2(dba)_3{\cdot}CHCl_3$, 8 mol% of P(*o*-tol)$_3$, and 1.5 mmol **2** were used. [c] 4 mol% of $Pd(PPh_3)_4$ was used instead of $Pd_2(dba)_3{\cdot}CHCl_3$ and P(*o*-tol)$_3$.

cyclobutenone (**5**) with 76% yield *via* a tandem Michael addition/cyclization reaction.[7b] Reduction by DIBAL-H could give tetrasubstituted allenol (**6**) with 51% yield. Hydrolysis by LiOH·H_2O afforded tetrasubstituted allenic acid (**7**) in 70% yield.[12e]



Scheme 2 Synthetic applications of tetrasubstituted 2,3-allenoates **3aa** and **3ba**.

A plausible mechanism is shown in Scheme 3: the allenylpalladium intermediate **I** generated from the S_N2'-type oxidative addition of a Pd(0) complex with the propargylic carbonates underwent transmetalation with arylboronic acids to give the allenylpalladium intermediate **II**, which resulted in the formation of 2,3-allenoates *via* reductive elimination.



Scheme 3 A proposed mechanism.

Finally, we were also intrigued by the potential biological activities of these newly synthesized tetrasubstituted 2,3-allenoates, and thus their cytotoxicity against A549 human lung cancer cells was evaluated. Cell growth was measured by the sulforhodamine B (SRB) assay. The inhibition effect of the compounds on cell growth was firstly screened at 10 μM, and then those with an inhibitory rate higher than 50% were further investigated and IC_{50}s (concentration to inhibit cell via-

Table 4 Cytotoxicity against A549 human lung cancer cells

Compound	IR[a] (%)	IC_{50}[b] (μm)
3aa	—	8.6 ± 2.0
3bb	—	8.9 ± 0.6
3ae	—	6.7 ± 0.7
3af	—	6.1 ± 0.8
3ag	—	7.0 ± 0.4
3ah	21.7	
3ai	40.8	
3aj	—	10.6 ± 2.5
3ak	21.4	
3al	—	5.2 ± 0.4
3dn	—	2.4 ± 0.1
3ea	—	5.1 ± 2.0
3fa	46.4	
4	18.2	
5	18.1	
7	3.3	
Taxol[c]	66.7	

[a] Inhibitory rate of growth at 10 μM (compounds **3aa–7**) or 25 nM (Taxol). [b] IC_{50}s are presented as mean ± SD. [c] Taxol was used as a positive control.

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bility by 50%) were measured. As shown in Table 4, Compounds **3ah**, **3ai**, **3ak**, **3fa**, **4**, **5**, and **7** displayed marginal activity against the growth of A549 cells, whereas the rest of the compounds exhibited significant cytotoxicity with IC_{50}s in the low micromolar range. Among them, **3dn** showed the most potent activity with an IC_{50} of 2.4 μM, suggesting that it might be a promising anticancer agent for further investigation.

We also tested the potential activity of these tetrasubstituted 2,3-allenoates against diabetes. Glucagon-like peptide-1 (GLP-1) plays a critical role in limiting postprandial blood glucose and thus becomes an important target for developing new therapies for type II diabetes. STC-1 mouse intestinal endocrine cells were treated with the tested compounds and the extracellular level of GLP-1 was determined. As shown in Table 5, **3aa**, **3ag**, **3aj**, **3ak**, **3dn**, and **3fa** effectively stimulated the secretion of GLP-1 at 100 μM, which was more than 90% of that in the presence of the positive control PMA (1 μM). Furthermore, these six compounds were tested at a range of lower concentrations and the half maximal effective concentration (EC_{50}) was measured. Notably, **3aa** and **3dn** were the most potent with an EC_{50} of ~3 μM. Further investigations of the potential antidiabetic activity and the mechanisms are underway.

Table 5 Stimulation of the secretion of GLP-1 by STC-1 cells

Compound	Relative activity[a] (%)	EC_{50} [b] (μm)
3aa	98.8	3.6 ± 1.5
3bb	20.3	
3ae	80.1	
3af	87.6	
3ag	98.9	10.4 ± 0.6
3ah	0.05	
3ai	89.8	
3aj	103.1	6.7 ± 1.7
3ak	116.7	8.0 ± 2.1
3al	74.0	
3dn	102.3	3.3 ± 1.0
3fa	102.3	6.5 ± 2.0
4	62.6	
5	64.6	
7	64.2	

[a] Stimulation of GLP-1 secretion by the tested compounds at 100 μM. Data are normalized to the level of GLP-1 in the presence of 1 μM PMA. [b] EC_{50}s are presented as mean ± SD.

Conclusions

In summary, we have presented a straightforward protocol to prepare tetrasubstituted 2,3-allenoates from a wide range of 3-alkoxycarbonyl propargylic carbonates and boronic acids *via* palladium-catalysis under mild conditions with good to excellent yields tolerating syntheticen useful functional groups. These highly functionalized allenoate products may be easily transformed to other useful molecules. Preliminary biological studies found that several tetrasubstituted 2,3-allenoates exhibited both anti-cancer and anti-diabetic activities. Further studies including enantioselective synthesis of tetrasubstituted 2,3-allenoates *via* Suzuki-coupling are being conducted in our laboratory.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

Financial support from the National Natural Science Foundation of China (Grant No. 21690063 for S. Ma), the National Basic Research Program (2015CB856600 for S. Ma), and the Shanghai Sailing Program (18YF1402000 for H. Qian) is greatly appreciated. We also thank Mr Zheng Yang in this group for reproducing the results of **3ag**, **3cm**, and **3bd** presented in Table 3.

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